

FPGA AI Suite 2026.1.1 Installation Guide

Printable guide for Windows 11 and Linux/Ubuntu users

Release focus: FPGA AI Suite 2026.1.1

OpenVINO requirement: OpenVINO Toolkit 2025.4

Recommended beginner install: Docker image

1. What this guide covers

This guide gives a practical installation and verification flow for:

1. Windows 11 systems using Docker Desktop.
2. Linux/Ubuntu systems using Docker Engine.
3. The complete FPGA AI Suite Docker image with Quartus included.
4. Basic verification with dla_compiler.
5. ResNet-50 model conversion, compilation, area/performance estimation, and software emulation.
6. Initial Agilex 5 E-Series Modular Development Kit notes.

The current complete image used in this guide is:

```
alterafpga/fpgaaisuite:2026.1.1-quartus
```

Use this image when you want FPGA AI Suite plus Quartus tools in one environment.

2. Important version support note

For FPGA AI Suite 2026.1.1, the official native Linux operating systems are:

```
Ubuntu 22.04 LTS
Ubuntu 24.04 LTS
Red Hat Enterprise Linux 8.10 / 9.4
```

Ubuntu 26.04 is not listed as an officially supported native installation target. On Ubuntu 26.04, the recommended practical path is Docker.

3. Key concepts

FPGA AI Suite components

Component	Purpose
OpenVINO	Converts and represents AI models in IR format
'xml' model file	OpenVINO model graph/layer structure
'bin' model file	OpenVINO model weights
'dla_compiler'	FPGA AI Suite compiler for area/performance/AOT output
'arch' file	FPGA AI IP architecture description
'aot' file	Ahead-of-time compiled model
Runtime	Runs inference through emulation, JTAG, PCIe, or SoC flow
Quartus	Builds/programs FPGA hardware bitstreams

Quartus vs OpenVINO runtime

Quartus builds/configures FPGA hardware. OpenVINO plus the FPGA AI Suite runtime runs inference. Quartus does not replace OpenVINO runtime.

Part A - Windows 11 installation flow

A1. Choose the correct Docker installer

For Intel i5/i7/i9 or AMD Ryzen Windows systems, choose:

```
Docker Desktop for Windows - AMD64
```

Do not choose Windows ARM64 unless the machine has an ARM CPU.

A2. Install Docker Desktop

1. Download Docker Desktop from Docker.
2. Install it with WSL 2 backend enabled if prompted.
3. Restart Windows.
4. Open PowerShell and test:

```
docker --version  
docker run hello-world
```

A3. Pull FPGA AI Suite image

For complete FPGA AI Suite plus Quartus:

```
docker pull alterafpga/fpgaaisuite:2026.1.1-quartus
```

For smaller compiler/emulation-only use:

```
docker pull alterafpga/fpgaaisuite:2026.1.1-standalone
```

A4. Create Windows workspace

```
mkdir C:\fpga-ai-work
```

A5. Start container

Complete image:

```
docker run -it --name fpga-ai-suite-2026 -v C:\fpga-ai-work:/workspace alteraf  
pga/fpgaaisuite:2026.1.1-quartus
```

Restart later:

```
docker start -ai fpga-ai-suite-2026
```

A6. How to start FPGA AI Suite each time on Windows

After Docker Desktop is running, open PowerShell and start the existing container:

```
docker start -ai fpga-ai-suite-2026
```

You are inside FPGA AI Suite when the prompt changes from a Windows prompt like:

```
PS C:\Users\name>
```

to a Linux container prompt like:

```
fpga_ai_suite@container_id:~$
```

Then go to your workspace and reload the local FPGA AI Suite work environment:

```
cd /workspace/quickstart
source ./coredla_work.sh
```

If the workspace has not been created yet, use the workspace setup section later in this guide.

To exit the container:

```
exit
```

Do not create a new container every day unless you intentionally want a clean environment. Use `docker start -ai` for daily work.

A7. Verify inside Docker

Prompt should look like:

```
fpga_ai_suite@container_id:~$
```

Run:

```
python3 -c "import openvino; print(openvino.__version__)"
echo $COREDLA_ROOT
dla_compiler --version
quartus_pgm --version
jtagconfig
```

Expected OpenVINO version begins with:

```
2025.4
```

If no FPGA board is connected, `jtagconfig` may show:

```
No JTAG hardware available
```

That is normal.

Part B - Linux/Ubuntu Docker installation flow

B1. Recommended for Ubuntu 26.04

Ubuntu 26.04 is not officially listed as a native FPGA AI Suite target. Use Docker.

B2. Install Docker Engine

```
sudo apt update
sudo apt install -y ca-certificates curl gnupg
sudo install -m 0755 -d /etc/apt/keyrings
```

```
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -  
o /etc/apt/keyrings/docker.gpg  
sudo chmod a+r /etc/apt/keyrings/docker.gpg
```

If Docker has a repo for your Ubuntu codename, use it. If not, use the Ubuntu 24.04 noble repo:

```
echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docke  
r.gpg] https://download.docker.com/linux/ubuntu noble stable" | sudo tee /etc/  
apt/sources.list.d/docker.list > /dev/null
```

Install Docker:

```
sudo apt update  
sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin  
docker-compose-plugin  
sudo systemctl enable docker  
sudo systemctl start docker  
sudo usermod -aG docker $USER
```

Log out and log back in. Test:

```
docker run hello-world
```

B3. Pull complete FPGA AI Suite image

```
docker pull alterafpga/fpgaaisuite:2026.1.1-quartus
```

B4. Create workspace

```
mkdir -p ~/fpga-ai-work
```

B5. Run container with USB/JTAG support

Use this command if you may connect an FPGA board later:

```
docker run -it \  
--name fpga-ai-suite-2026 \  
--privileged \  
-v /dev/bus/usb:/dev/bus/usb \  
-v ~/fpga-ai-work:/workspace \  
alterafpga/fpgaaisuite:2026.1.1-quartus
```

Restart later:

```
docker start -ai fpga-ai-suite-2026
```

B6. How to start FPGA AI Suite each time on Linux

Open a terminal and start the existing container:

```
docker start -ai fpga-ai-suite-2026
```

You are inside FPGA AI Suite when the prompt changes from the host prompt, for example:

```
sahilpat@sahilpat:~$
```

to the container prompt, for example:

```
fpga_ai_suite@container_id:~$
```

Reload your working directory environment:

```
cd /workspace/quickstart
source ./coredla_work.sh
```

If you need USB-JTAG access and the original container was not created with USB pass-through, recreate it with `--privileged -v /dev/bus/usb:/dev/bus/usb`.

To exit the container:

```
exit
```

If you need to recreate the container:

```
docker rm fpga-ai-suite-2026
```

B7. Verify inside Docker

```
python3 -c "import openvino; print(openvino.__version__)"
echo $COREDLA_ROOT
dla_compiler --version
quartus_pgm --version
jtagconfig
```

Expected examples:

```
OpenVINO: 2025.4.x
FPGA AI Suite: 2026.1.1
Quartus Programmer: 26.1
```

Part C - Workspace setup

C1. Create FPGA AI Suite working directory

Inside Docker:

```
cd /workspace
mkdir -p quickstart
cd quickstart
source dla_init_local_directory.sh
```

If already initialized:

```
source ./coredla_work.sh
```

Verify:

```
echo $COREDLA_WORK
```

Expected:

```
/workspace/quickstart
```

C2. Fix workspace permission issue if needed

If you see:

```
mkdir: cannot create directory: Permission denied
```

Inside Docker:

```
sudo chown -R fpga_ai_suite:fpga_ai_suite /workspace
```

Then retry workspace creation.

Part D - Verify compiler and architecture files

D1. Check Agilex 5 architecture files

```
ls $COREDLA_ROOT/example_architectures | grep AGX5
```

Common files include:

```
AGX5_Generic.arch  
AGX5_Performance.arch  
AGX5_FP16_Generic.arch  
AGX5_Small_NoSoftmax.arch
```

D2. Run area estimate

For Agilex 5:

```
dla_compiler --fanalyze-area --march $COREDLA_ROOT/example_architectures/AGX5_  
Generic.arch
```

Typical report contains:

```
ALMs  
ALUTs  
DSPs  
Registers  
M20Ks  
Memory ALMs
```

Part E - ResNet-50 example flow

E1. Download ResNet-50

```
cd $COREDLA_WORK  
omz_downloader --name resnet-50-tf --output_dir $COREDLA_WORK/demo/models/
```

E2. Convert to OpenVINO IR

```
omz_converter --name resnet-50-tf \  
--download_dir $COREDLA_WORK/demo/models/ \  
--output_dir $COREDLA_WORK/demo/models/
```

Verify:

```
ls $COREDLA_WORK/demo/models/public/resnet-50-tf/FP32/
```

Expected:

```
resnet-50-tf.xml
resnet-50-tf.bin
```

E3. Compile for Agilex 5

```
cd $COREDLA_WORK/demo/models/public/resnet-50-tf/FP32

dla_compiler \
  --march $COREDLA_ROOT/example_architectures/AGX5_Generic.arch \
  --network-file ./resnet-50-tf.xml \
  --o $COREDLA_WORK/demo/RN50_AGX5_Generic_b1.aot \
  --fanalyze-performance
```

Success line:

```
Performance Estimate has finished for Neural Network 1
```

Check AOT file:

```
ls -lh $COREDLA_WORK/demo/RN50_AGX5_Generic_b1.aot
```

E4. View performance report

From the model FP32 directory:

```
cat compiled_model_dir/TensorFlow_Frontend_IR/reports/performance-report_0.txt
```

Key value:

```
FINAL THROUGHPUT
```

Part F - Software emulation flow

F1. Build runtime for emulation

```
cd $COREDLA_WORK/runtime
./build_runtime.sh --target_emulation
```

F2. Set variables

```
model_dir=$COREDLA_WORK/demo/models/public
image_dir=$COREDLA_WORK/demo/sample_images
cur_arch=$COREDLA_ROOT/example_architectures/AGX5_Generic.arch
```

F3. Run emulator inference

```
cd $COREDLA_WORK/runtime/build_Release/dla_benchmark

./dla_benchmark \
  -b 1 \
```

```
-niter 1 \
-nireq 1 \
-m $modeldir/resnet-50-tf/FP32/resnet-50-tf.xml \
-d HETERO:FPGA,CPU \
-i $imagedir \
-arch_file $curarch \
-dump_output \
-plugins emulation \
-bgr
```

Success line:

PASSED

Emulation is slow. It is for correctness/debugging, not performance.

Part G - Agilex 5 E-Series Modular Development Kit

G1. Your board

Board:

Agilex 5 FPGA and SoC E-Series Modular Development Kit
MK-A5E065BB32AES1

Relevant FPGA AI Suite design identifiers:

Design type	Identifier
OFs PCIe-attached	'agx5e_modular_ofs_pcie'
Hostless JTAG-attached	'agx5e_modular_jtag'
Hostless Spatial JTAG	'agx5e_modular_spatial_jtag'
On-chip parameter DDR-free	'agx5_modular_ddrfree'
DDR-overflow	'agx5_modular_ddrfree_ltdrwb'
SoC M2M	'agx5_soc_m2m'
SoC S2M	'agx5_soc_s2m'

G2. Recommended first board flow

Start with:

Hostless JTAG-attached
Identifier: agx5e_modular_jtag
Architecture: AGX5_Generic.arch

G3. Build JTAG runtime

```
cd $COREDLA_WORK/runtime
./build_runtime.sh -target_system_console
```

G4. Build Agilex 5 JTAG bitstream

```
cd $COREDLA_WORK
$COREDLA_ROOT/bin/dla_build_example_design.py build \
--licensed \
--output-dir build_agx5_jtag_ed \
--num-instances 1 \
--seed 1 \
```

```
agx5e_modular_jtag \
$COREDLA_ROOT/example_architectures/AGX5_Generic.arch
```

Expected output:

```
$COREDLA_WORK/build_agx5_jtag_ed/AGX5_Generic.sof
```

G5. Program board

Connect board power and USB-JTAG. Then:

```
jtagconfig
jtagconfig --setparam 1 JtagClock 16M
cd $COREDLA_WORK/build_agx5_jtag_ed
quartus_pgm -c 1 -m jtag -o "p;AGX5_Generic.sof"
```

JTAG is slow. Use `-nireq=1` for JTAG examples.

G6. Permanent licensed Docker startup

For a node-locked Altera license, start Docker with host networking, USB pass-through, workspace mount, license mount, and license environment variables.

Create or update this host-side script:

```
mkdir -p ~/bin
nano ~/bin/start-fpga-ai-suite.sh
```

Use this content, adjusting the license filename if needed:

```
#!/bin/bash
set -e

CONTAINER_NAME="fpga-ai-suite-2026"
IMAGE_NAME="alterafpga/fpgaaisuite:2026.1.1-quartus"
WORKSPACE="$HOME/fpga-ai-work"
LICENSE_FILE="$HOME/altera_pro/LR-178044_License.dat"

if [ ! -f "$LICENSE_FILE" ]; then
    echo "ERROR: License file not found: $LICENSE_FILE"
    exit 1
fi

mkdir -p "$WORKSPACE"

if docker ps -a --format '{{.Names}}' | grep -qx "$CONTAINER_NAME"; then
    echo "Starting existing FPGA AI Suite container..."
    docker start -ai "$CONTAINER_NAME"
else
    echo "Creating new licensed FPGA AI Suite container..."
    docker run -it \
        --name "$CONTAINER_NAME" \
        --network host \
        --privileged \
        -v /dev/bus/usb:/dev/bus/usb \
        -v "$WORKSPACE":/workspace \
        -v "$LICENSE_FILE":/licenses/altera_license.dat:ro \
        -e LM_LICENSE_FILE=/licenses/altera_license.dat \
        -e ALTERAD_LICENSE_FILE=/licenses/altera_license.dat \
        "$IMAGE_NAME"
```

```
fi
```

Make it executable:

```
chmod +x ~/bin/start-fpga-ai-suite.sh
```

Daily startup:

```
~/bin/start-fpga-ai-suite.sh
```

Inside Docker, verify the license variables and tools:

```
echo $LM_LICENSE_FILE
echo $ALTERAD_LICENSE_FILE
ls -lh /licenses/altera_license.dat
which quartus_pgm
which system-console
which lmutil
```

If system-console or lmutil is not in PATH, add:

```
echo 'export PATH=/opt/altera/syscon/bin:/opt/altera/qcore/linux64:$PATH' >> ~/.bashrc
echo 'export QUARTUS_ROOTDIR=/opt/altera/quartus' >> ~/.bashrc
source ~/.bashrc
```

Part H - Object detection notes

FPGA AI Suite includes ported OpenVINO demo applications:

```
classification_sample_async
object_detection_demo_yolov3_async
segmentation_demo
```

The object detection demo is documented under the PCIe design example section and uses YOLOv3:

```
yolo-v3-tf
yolo-v3-tiny-tf
```

The sample command in the handbook is shown for a PCIe/Agilex 7 example. For the Agilex 5 E-Series Modular Development Kit, the relevant full application-style path is closer to:

```
agx5e_modular_ofs_pcie
```

JTAG can run inference, but it is not ideal for live/video object detection because USB-JTAG is slow.

H1. Download and compile YOLOv3 for AGX5

```
cd $COREDLA_WORK
omz_downloader --name yolo-v3-tf --output_dir $COREDLA_WORK/demo/models/

omz_converter --name yolo-v3-tf \
  --download_dir $COREDLA_WORK/demo/models/ \
  --output_dir $COREDLA_WORK/demo/models/

cd $COREDLA_WORK/demo/models/public/yolo-v3-tf/FP32
```

```
dla_compiler \  
--march $COREDLA_ROOT/example_architectures/AGX5_Generic.arch \  
--network-file ./yolo-v3-tf.xml \  
--o $COREDLA_WORK/demo/YOLOv3_AGX5_Generic_b1.aot \  
--fanalyze-performance
```

Part I - Troubleshooting

I1. Docker container name conflict

Error:

```
container name is already in use
```

Fix:

```
docker start -ai fpga-ai-suite-2026
```

or recreate:

```
docker rm fpga-ai-suite-2026
```

I2. OpenVINO not found

If this fails on host Ubuntu:

```
python3 -c "import openvino; print(openvino.__version__)"
```

that is expected if OpenVINO is only installed inside Docker. Enter container first:

```
docker start -ai fpga-ai-suite-2026
```

I3. Workspace permission denied

Inside Docker:

```
sudo chown -R fpga_ai_suite:fpga_ai_suite /workspace
```

I4. License issue: No valid license for CoreDLA found

Symptoms in build.log:

```
No valid license for CoreDLA found.  
Building unlicensed version.
```

First confirm the license file is mounted inside Docker:

```
echo $LM_LICENSE_FILE  
echo $ALTERAD_LICENSE_FILE  
ls -lh /licenses/altera_license.dat
```

Find lmutil if needed:

```
find /opt/altera -name lmutil 2>/dev/null
```

In the Quartus Docker image used here, lmutil was located at:

```
/opt/altera/qcore/linux64/lmutil
```

Add it to PATH:

```
export PATH=/opt/altera/qcore/linux64:$PATH
```

Check the container-visible host ID:

```
lmutil lmhostid
```

Check the FPGA AI Suite/CoreDLA features from the license. The useful feature IDs observed for FPGA AI Suite were:

```
6AF8_018B
6AF7_018B
```

Run diagnostics:

```
lmutil lmdiag -c /licenses/altera_license.dat 6AF8_018B
lmutil lmdiag -c /licenses/altera_license.dat 6AF7_018B
```

Success looks like:

```
This is the correct node for this node-locked license
```

If you see:

```
Invalid host
```

then the license is locked to a different NIC/host ID. Generate or rehost the license using the real Linux host Ethernet MAC, not the Docker virtual MAC. Confirm host MACs on the Linux host with:

```
ip link
```

For this setup, the corrected license was generated for primary computer ID:

```
e89744bdaefd
```

For node-locked licenses in Docker, start the container with:

```
--network host
```

so FlexLM can see the host network identity.

For the Agilex 5 JTAG design build, force the licensed IP path with:

```
$COREDLA_ROOT/bin/dla_build_example_design.py build \
--licensed \
--output-dir build_agx5_jtag_ed \
--num-instances 1 \
--seed 1 \
agx5e_modular_jtag \
$COREDLA_ROOT/example_architectures/AGX5_Generic.arch
```

The build log should include:

```
Created licensed IP.
Building licensed version.
```

I5. No JTAG hardware available

Possible causes:

1. Board not powered.
2. USB-JTAG cable not connected.
3. Container not started with USB pass-through.
4. Host USB permissions issue.

Use container command with:

```
--privileged -v /dev/bus/usb:/dev/bus/usb
```

I6. Accuracy check fails but emulator says PASSED

If emulator prints:

```
PASSED
```

then emulation ran. Accuracy mismatch usually means preprocessing or ground-truth mismatch. Try first without `-groundtruth_loc`, then inspect output files.

Final checklist

A complete working environment should pass:

```
python3 -c "import openvino; print(openvino.__version__)"
echo $COREDLA_ROOT
dla_compiler --version
quartus_pgm --version
ls $COREDLA_ROOT/example_architectures | grep AGX5
dla_compiler --fanalyze-area --march $COREDLA_ROOT/example_architectures/AGX5_
Generic.arch
```

A successful model flow should produce:

```
resnet-50-tf.xml
resnet-50-tf.bin
RN50_AGX5_Generic_b1.aot
Performance Estimate has finished
PASSED from emulator
```